

Self-Targeting in U.S. Transfer Programs

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Motivation

- Millions of Americans are eligible for means-tested transfers but don't claim them
 - Take-up rates: 65% for SNAP, 60% for Medicaid, 50% for SSI, ...
 - Key distinction between welfare states (U.S. versus Europe)
 - Relevant now: 7 million Medicaid-eligible Americans set to lose their benefits with the end of pandemic-era automatic enrollment, because of non-take-up
- Policy alternative to status quo: send help automatically, not upon application
 - Social benefit of this alternative: eliminate **ordeal costs** for recipients
 - Social cost: eliminate "**self-targeting**" via incomplete take-up among the eligible
- **This paper:** measure and evaluate this trade-off in eight U.S. transfer programs
 - ① *Measurement:* How self-targeted is transfer take-up with respect to need?
 - ② *Welfare:* Does the social value of self-targeting exceed the social cost of ordeals?

Motivation

Theory: Should transfer programs be voluntary or automatic?

- Advantageous self-targeting = necessary condition for voluntary transfers
- **Classic PF viewpoint:** ordeals → self-targeting (Nichols & Zeckhauser 1982)
- **Counterpoint:** ordeals can perversely screen out neediest (Currie & Gahvari 2008)

Empirics: Do ordeals induce advantageous self-targeting on average?

- Mixed literature on ordeals measuring **selection on the margin** →
 - Needed for voluntary vs. automatic: **selection on average among the eligible**
 - Why? Automatic transfer also redistributes to voluntary regime's "never-takers"
- **At stake:** voluntarism versus universality as basic principles of safety net design

This Paper

① How much self-targeting in U.S. transfer programs?

- Self-targeting is **advantageous** on consumption & lifetime income across 8 transfers
- Example: Average SNAP recipient consumes \$11,000 less per year (↓ 19 percentiles) than average **eligible nonrecipient with the same income**
- Automatic versus voluntary: Lowest-consumption HHs receive **50%–75% more under status quo** than under budget-neutral automatic program

② Should transfers be voluntary or automatic?

- Derive sufficient-statistics formulas for nonlinear tax/transfer with self-targeting
- Social benefit of self-targeting (model-based measure): **6 cents per transfer dollar**
- **Social benefit exceeds social cost** on average, but nuanced heterogeneity by program

→ **Conclusion:** self-targeting provides a compelling case for voluntary transfers

Outline

- ① Measurement
- ② Welfare Analysis

Data and Measurement

- **Sample:** PSID 1997–2019, heads of household & partners (age 18 to 65)
- **Current Income / Lifetime Income / Consumption:**
 - Equivalize for household size & composition (Citro & Michael 1995)
 - Impute consumption flow from home and car ownership (Meyer & Sullivan 2023)
 - Estimate lifetime income from incomplete panels (Haider & Solon 2006) →
- **Transfers:** Receipt for 8 consolidated programs (\$830B expenditure in 2019)

SNAP
Medicaid

Housing Assistance
SSI

TANF
WIC

LIHEAP
School Meals

Eligibility: new, detailed imputation code from state-by-year transfer rules →

Empirical Definition of Self-Targeting

A transfer is **self-targeting** on an outcome C_i if

$$\underbrace{E[C_i | D_i = E_i = 1, \mathbf{X}_i]}_{\text{recipient average}} < \underbrace{E[C_i | D_i = 0, E_i = 1, \mathbf{X}_i]}_{\text{eligible nonrecipient average}}$$

Emerges from “**correlation test**” framework (Chiappori & Salanié 2000):

$$\text{Consumption: } C_i = \mathbf{X}_i\delta + \alpha D_i + \beta E_i + \nu_i$$

$$\text{Take-Up: } D_i = \begin{cases} \mathbf{X}_i\gamma + \varepsilon_i & \text{if } E_i = 1 \\ 0 & \text{if } E_i = 0 \end{cases}$$

$$\text{Self-targeting condition: } \text{corr}(\nu_i, \varepsilon_i) < 0$$

Fact 1: Transfer Receipt Falls in Consumption Given Income

Transfer Dollars Per Capita		Income Quintile					Avg.
		1	2	3	4	5	
Consumption Quintile	1						
	2						
	3						
	4						
	5						
Avg.							

Fact 1: Transfer Receipt Falls in Consumption Given Income

Transfer Dollars Per Capita		Income Quintile					Avg.
		1	2	3	4	5	
Consumption Quintile	1	3,647	1,353	600	397	155	2,440
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	3						
	4						
	5						
Avg.							

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	2	1,745					666
	3	920					303
	4	572					153
	5	557					101
Avg.		2,435	844	266	92	27	

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	3	920	563	217	102	33	303
	4	572	403	168	60	33	153
	5	557	273	133	58	18	101
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Fact 2: Selection Driven by Take-Up, Not Eligibility

SNAP Eligibility Rate		Income Quintile					Avg.
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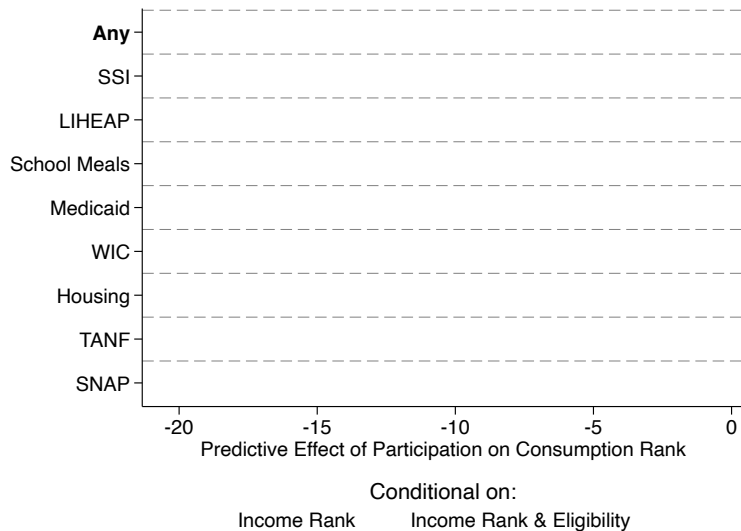
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SNAP Eligibility Rate		Income Quintile					Avg.
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Consumption Quintile	1	83.8	23.7	0.4	0.8	0.0	51.9
	2	75.5	15.8	0.4	0.2	0.0	19.0
	3	67.5	14.0	0.4	0.2	0.1	10.5
	4	61.3	13.9	0.4	0.3	0.1	7.2
	5	60.7	17.6	0.5	0.3	0.0	6.5
Avg.		76.3	18.1	0.4	0.3	0.1	
SNAP Take-Up Rate		Income Quintile					Avg.
		1	2	3	4	5	
Consumption Quintile	1	52.2	39.3	.	.	.	50.2
	2	26.8	23.7	.	.	.	25.9
	3	14.9	14.4	.	.	.	14.5
	4	8.3	8.2	.	.	.	8.1
	5	8.1	8.6	.	.	.	8.1
Avg.		37.5	27.2	.	.	.	

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Receipt Rate							

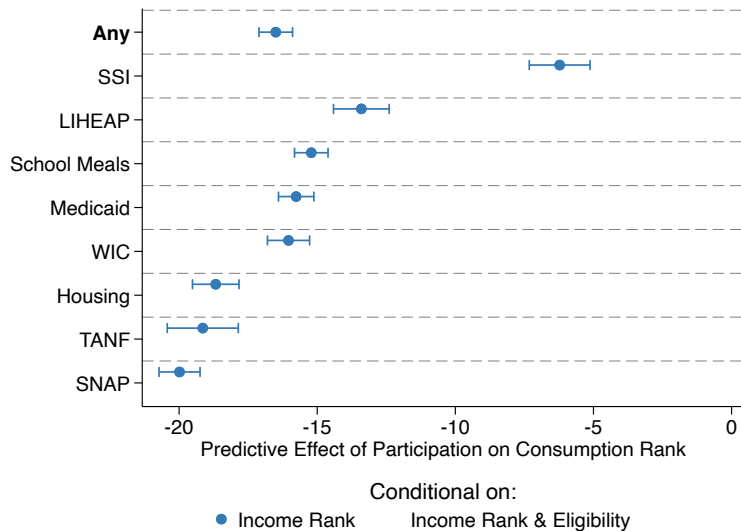
Fact 3: Self-Targeting Is Advantageous Across U.S. Transfer Programs



Lifetime Income

+ Controls

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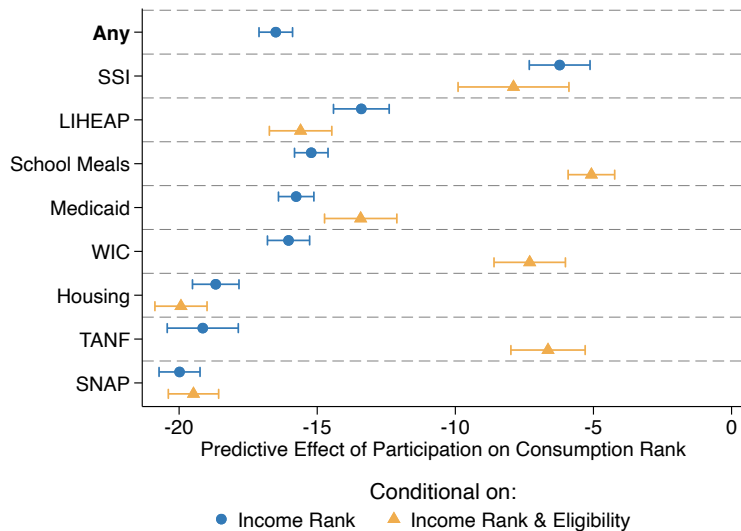


$$C_{it} = \beta D_{it} + f(Y_{it}) + u_{it}$$

Lifetime Income

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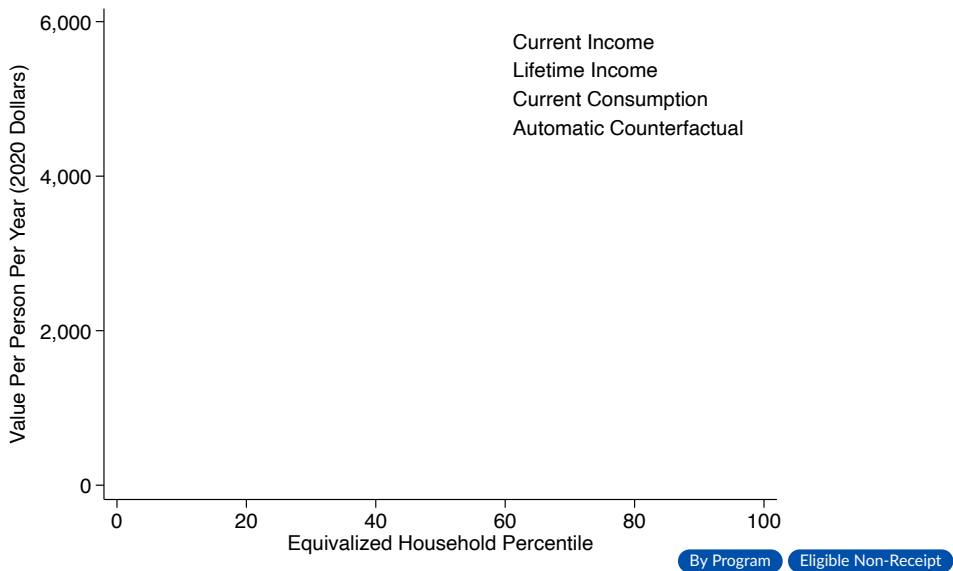
Among $E_{it} = 1$:

$$C_{it} = \beta D_{it} + f(Y_{it}) + u_{it}$$

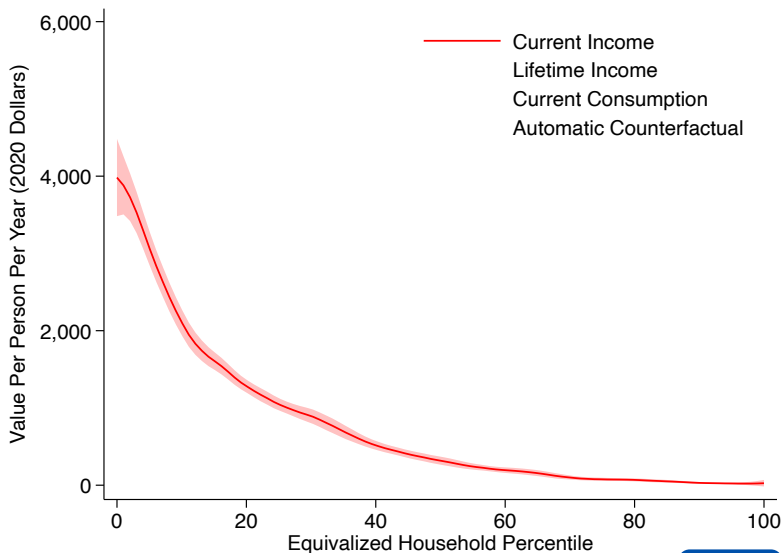
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Fact 4: Self-Targeting Raises Transfer Progressivity in Consumption



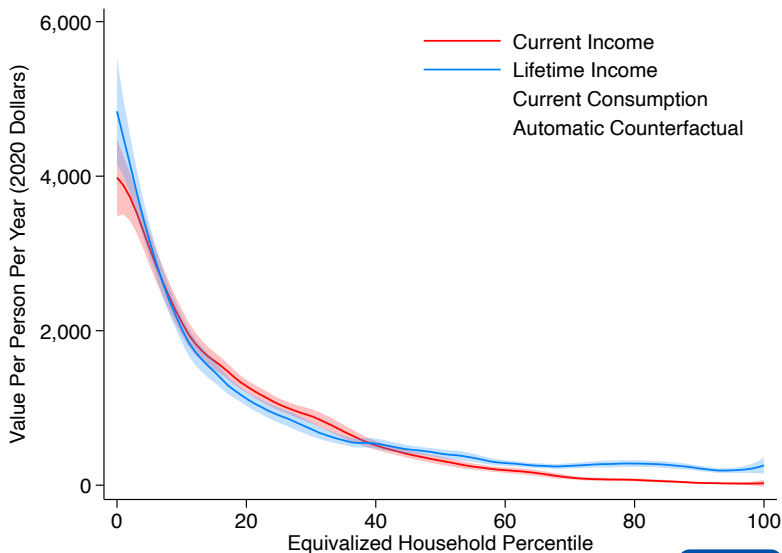
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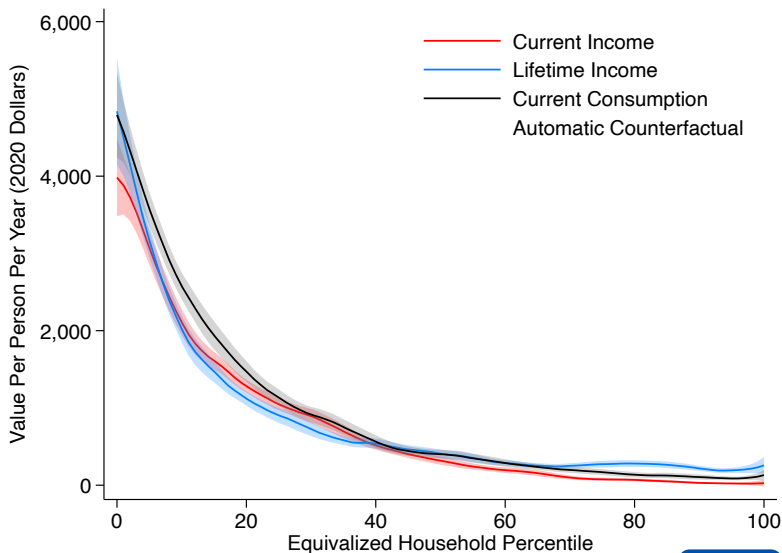
By Program

Eligible Non-Receipt

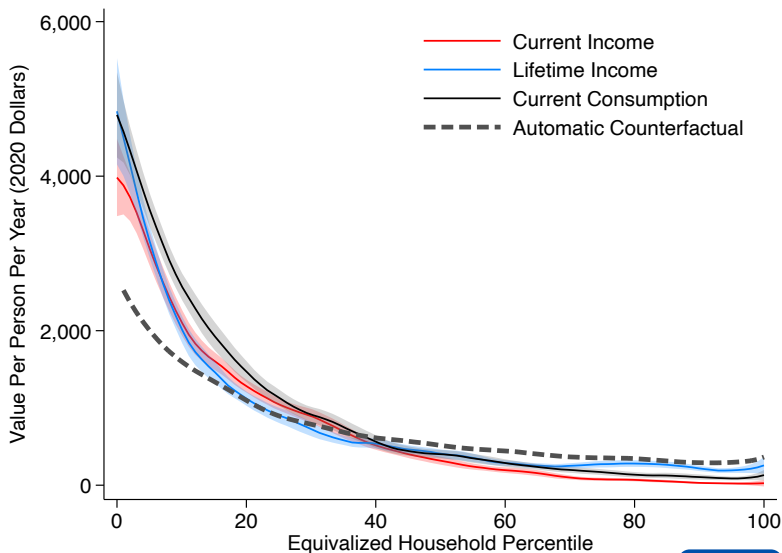
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Sensitivity to Mismeasurement

Key threat: [survey data quality](#) → address with thorough sensitivity analysis

- **Transfer receipt:** misreporting corrections [raise self-targeting](#) (Mittag 2019) →
- **Eligibility:** Is self-targeting actually unobservable eligibility rules? [Probably not](#)
 - Results are robust to reclassifying simulated-ineligible recipients as eligible →
 - Find self-targeting even in demographic cells with near-complete eligibility →
- **Consumption:** Advantageous self-targeting of transfers holds for...
 - Ownership of consumer durable goods (Meyer & Sullivan 2012) →
 - “Well-measured” consumption sub-categories (Meyer & Sullivan 2023) →

More sensitivity: [Current Income](#) [Lifetime Income](#) [Welfare Dependence](#)

Outline

- ① Measurement
- ② Welfare Analysis

Welfare Analysis: Motivating Example

Reform: cut \$1 from a voluntary transfer & split fiscal savings via an automatic transfer

- 100 people: 50 get voluntary transfer (\$ B); 49 inframarginal takers, 1 marginal
- Welfare weights: α_{AT} for inframarginals $\geq \alpha_C$ for indifferent $\geq \alpha_{NT}$ for non-takers

$$\begin{aligned}\Delta W = & \underbrace{49 \times \alpha_{AT} \times \left(\frac{49+B}{100} - 1 \right)}_{\text{Welfare impact on inframarginal takers}} + \underbrace{50 \times \alpha_{NT} \times \frac{49+B}{100}}_{\text{Welfare impact on inframarginal non-takers}} \\ & + \underbrace{1 \times \alpha_C \times \frac{49+B}{100}}_{\text{Welfare impact on marginals}}\end{aligned}$$

→ Welfare is reduced if $\alpha_{AT} \gg \alpha_{NT}$ (**self-targeting benefit**) and B is small (**ordeal cost**)

→ Intuition carries through into optimal nonlinear tax/transfer model

Welfare Analysis: Sufficient Statistics Formula

Welfare impact of reallocating ds from voluntary to automatic transfer:

$$dW = \underbrace{ds \int_z M(z) (E_{\kappa} [\alpha(z, \kappa)] - E_{\kappa \leq S(z)} [\alpha(z, \kappa)]) h(z) dz}_{\text{lost social benefit from self-targeting}} \\ + \underbrace{E_z[S(z)] m(z) E_{z, \kappa} [\alpha(z, \kappa)]}_{\text{social savings on ordeals}} + \text{labor supply effects}$$

- $S(z)$: value of voluntary transfer at income z
- $M(z)$: voluntary transfer receipt rate at income z ($m(z) = dM(z)/dS(z)$)
- $h(z)$: density of income at z
- $\alpha(z, \kappa)$: social welfare weight at income z and idiosyncratic take-up cost κ

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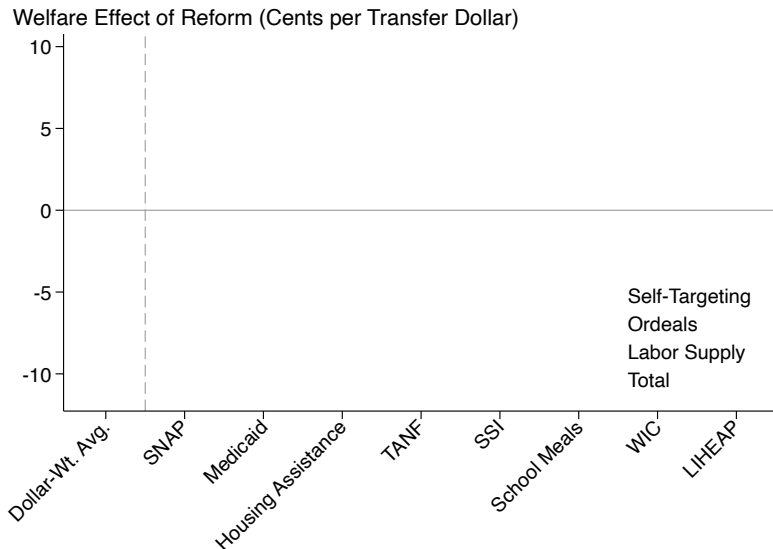
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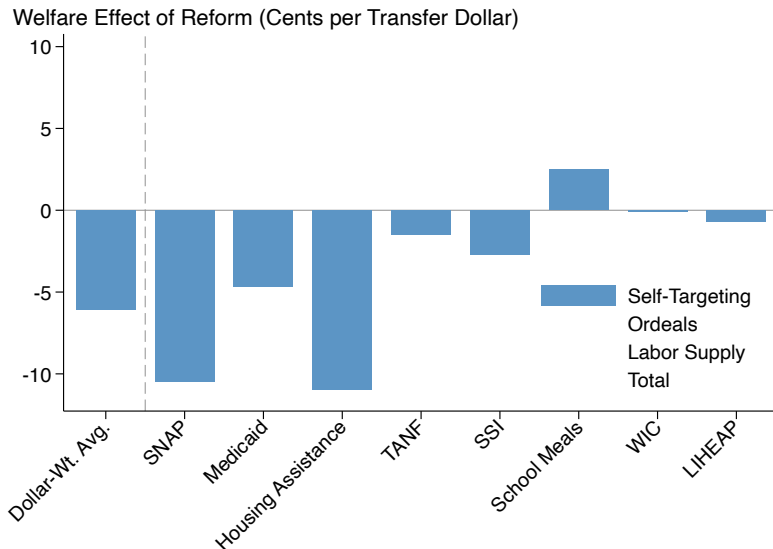
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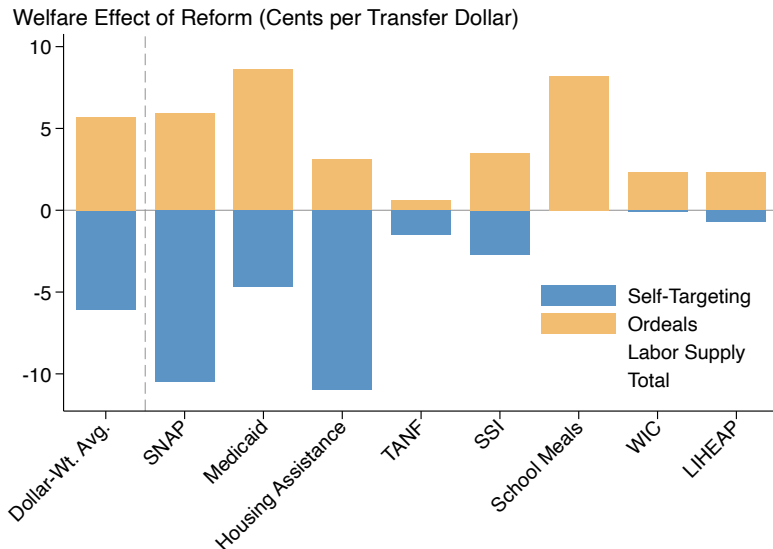
Welfare Analysis: Quantification



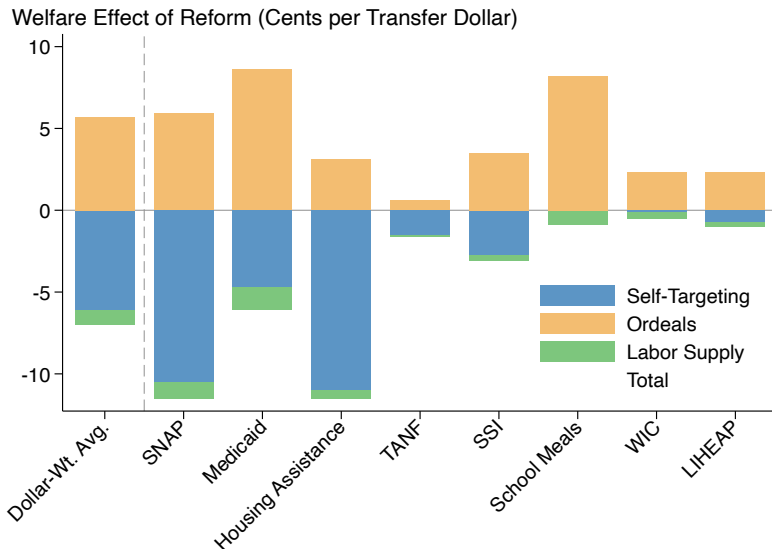
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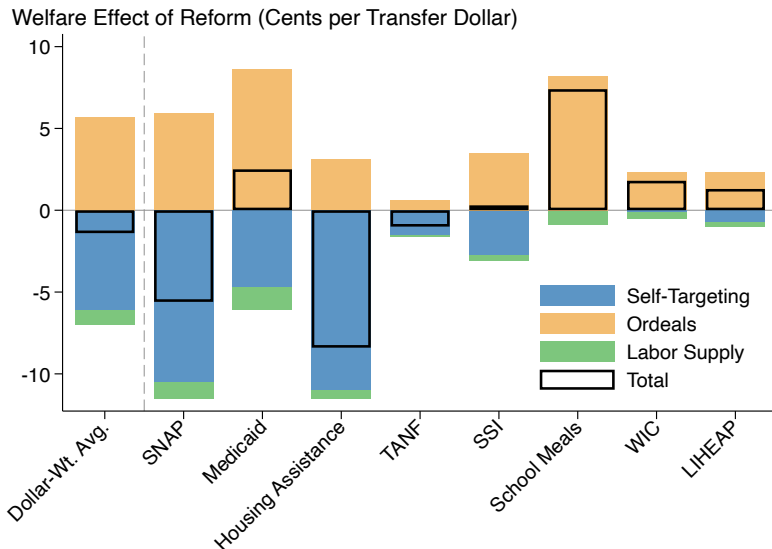
Welfare Analysis: Quantification



Welfare Analysis: Quantification



Welfare Analysis: Quantification



Conclusion

This paper: Should transfer programs be voluntary or automatic?

- Fundamental question for the design of social safety nets
- Renewed interest amid post-pandemic pullback of transfers in U.S.
- **Core trade-off:** Social benefit of self-targeting versus social cost of ordeals

Our answer: Benefits of self-targeting likely exceed ordeal costs

- **Measurement:** Document advantageous self-targeting in eight U.S. transfers
- **Welfare:** Quantify trade-off using sufficient-statistics approach

Thank you!

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